

The importance of city innovation networks in regional economic complexity

One of the most interesting topics for me, involving both small-world networks and economic complexity are city innovation networks and their utter importance for the economic complexity of regions.

In more modern economies, cities are the most vital parts of innovations, as well as productivity. However, we often experience that cities do not perform equally in this regard: Some of them fall behind when it comes to innovation and productivity, while others thrive and flourish. One of the most popular explanations for this phenomenon lies within the structure of urban collaborations, which creates citywide small-world networks that combine both local clustering and global connectivity. The understanding of how these networks can affect regional economic complexity is crucial in order to design impactful and effective policies in both urban and regional context to promote sustainable growth.

Recent research has highlighted the strong relationship between network connectivity and the complexity of an economy. One of our first papers in this course, Hidalgo and Hausmann (2009) showed that complex economies emerge from more intricate and densely interconnected production networks. Balland et al. (2018) extended this to regions and larger cities, finding that diversification and innovation depend on relatedness across industries, later also showing the connection of increasing inequality between cities to the complexity of today's economy (Balland et al. (2020)).

With these results and several questions answered, there are still gaps to be filled in to fully understand the underlying mechanics of economic complexity. One of the parts of this area that is seldom looked over include the understanding of how multiple layers of these urban networks interact with each other and how these synergies affect economic complexity and the growth in innovation. Since modern cities are now able to collaborate both physically and digitally, several new channels and networks can be explored with new ways to channel and circulate innovation. A related question can be formulated in the following way: *How do multi-layered (digital and local ties) urban network collaborations influence innovation growth? How does it influence the growth of economic complexity?*

Stakeholders can also benefit from uncovering this vital part of the topic. With this understanding, urban policymakers, for example, would be able to drive cities towards being more connected to several sectors of the economy through their interventions, helping to mitigate inequality between cities. Universities and research institutions could also benefit from the collaborations through regions, being able to spread out their network, creating new partnerships. Government agencies can also gain from this knowledge, since one of their main goals is supporting innovation and diversification.

References:

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